

Technical Review of the Liulan Reservoir “7·6” Dam Failure Disaster in Hengzhou, Guangxi, China

— Overtopping and Local Breach Development under Extreme Rainfall and Above-Standard Flooding: Processes and Broader Lessons

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Note: This brief is an independent technical review of the Liulan Reservoir “7·6” dam failure disaster, based on State Council announcements, public reporting, engineering-document leads, pre- and post-event imagery, and professional discussion available through July 25, 2026. Data and mechanistic interpretations not yet confirmed by original project records or the formal investigation are explicitly identified as uncertain. This brief does not replace the State Council investigation and assessment, engineering safety appraisal, hydrologic reassessment, or determination of responsibility. The title and the term “dam failure disaster” follow the official designation used by the State Council investigation team; the process description continues to distinguish overtopping, local breach initiation, and breach enlargement.

Abstract

On July 6, 2026, Liulan Reservoir in Hengzhou, Nanning, Guangxi, China, experienced overtopping, local breach initiation, and rapid breach enlargement during typhoon-related extreme rainfall and rapidly increasing reservoir inflow. On July 24, the State Council established an investigation and assessment team for the Liulan Reservoir “7·6” dam failure disaster. Based on publicly available reports, the State Council announcement, pre- and post-event imagery, and professional discussions available through July 25, this brief provides a phased technical review of the project background, rainfall and inflow, reservoir operation, actual gate performance, emergency spillway facilities, overtopping and breach development, downstream impacts, and post-disaster recovery. Because the original hydrologic, gate-operation, design, and monitoring records have not yet been fully released, the numerical values and mechanistic interpretations presented here remain subject to confirmation by the formal investigation.

Public information indicates that staged gate releases began on the evening of July 5. When operators planned to fully open all five gates early on July 6, some gates and hoisting equipment did not reach the intended fully open position. The fuse-plug embankment at the entrance to the emergency spillway on the east side did not breach as intended and could not be activated manually in time; its exact design conditions and the reasons it did not activate remain under investigation.

Operations were also constrained by the downstream safe release rate, evacuation progress, communications failures, and insufficient hydrologic data. The available evidence more strongly supports a staged sequence in which the reservoir water level reached the dam crest and overtopping began; concentrated erosion and an early breach then developed at the eastern abutment almost simultaneously or shortly thereafter; and a breach in the central portion of the dam subsequently formed and enlarged. At present, it is not possible to determine precisely whether a clearly distinguishable time interval existed between the onset of overtopping and initial breaching at the east abutment.

The combined evidence indicates that extreme hydrologic loading was the primary external driver of the event. The available information supports the view that erosion of the homogeneous earthfill dam under sustained overtopping was likely one of the key mechanisms responsible for local breach initiation and rapid enlargement. Gate reliability, emergency spillway performance, local reservoir geometry and possible flow concentration near the dam, monitoring and forecasting, operational decisions, communications, and downstream exposure all influenced the risk chain. The event cannot be attributed to a single factor. The dam-safety community should both reassess design floods, discharge capacity, and engineering protection standards under nonstationary climate conditions and manage the residual risk that remains when design standards are exceeded. Dam safety should be evaluated on the basis of the current condition and remaining safety margin of the entire system, rather than the year of construction alone.

1. Project Overview and Functional Role: A Profile of Liulan Reservoir

Liulan Reservoir is located in Liulan Village, Xiaoyi Town, Hengzhou City, Nanning, Guangxi, China, and impounds the Zhenlong River (also known as the Yunbiao River), a first-order tributary of the Yu River. Public sources report a drainage area of approximately 195 km² above the dam and a total storage capacity of about 93.19 million m³, although another published figure is approximately 95.52 million m³. The maximum dam height is about 42 m. The main dam is a homogeneous earthfill embankment constructed from 1958 to 1960. It is classified as a medium-sized reservoir serving irrigation, water supply, flood control, hydropower, ecological, and fishery functions. The principal publicly reported parameters are summarized in Appendix 1 (Appendix Table 1).

A homogeneous earthfill dam is constructed by placing and compacting cohesive soil in layers. Its safety cannot be judged simply by whether it is “new” or “old.” Unless specifically designed or protected for overtopping, an earthfill dam—regardless of age—generally cannot safely withstand sustained, high-volume flow over the crest. The more important implication of long service life is that the embankment materials, foundation, spillways, gate-hoisting systems, hydrologic design basis, monitoring facilities, maintenance capacity, and management arrangements may deteriorate or become inadequate at different rates.

Public reports indicate that the reservoir had previously experienced maintenance-related problems, including slope instability, seepage, and cracking, and that safety-rehabilitation works and standardized management upgrades had been implemented. Public interviews also indicate that a safety appraisal around 2020 generally concluded that the project was fit for safe operation, while noting problems involving rainfall and flood reporting, monitoring facilities, metal structures, or supporting management systems. These findings are not contradictory: the conclusion of a single appraisal does not mean that every component will thereafter maintain the same reliability, nor can it substitute for functional testing under extreme loading.

Liulan Reservoir irrigates approximately 157,300 mu (about 10,487 ha) of farmland and also supports water supply to nearby communities and a heavily funded irrigation-district modernization program. Public reporting indicates that the Yunbiao and Xiaoyi areas downstream are major jasmine-producing districts in Hengzhou, and that the flood damaged villages, flower fields, agricultural-supply businesses, roads, and water-supply facilities. Post-disaster decisions therefore should not be understood simply as “repairing the breach,” nor should they assume in advance that the original project must be reconstructed in the same form. Options should be compared in terms of functional recovery, alternative water supplies, risk reduction, and long-term operation and maintenance.

2. Disaster Process: Extreme Inflow, Effective Discharge Capacity, and Earthfill-Dam Overtopping

2.1 Extreme Rainfall, Inflow Magnitude, and Pre-Event Storage

The event occurred under Typhoon “Maysak”-related extreme rainfall. Early public reports indicated that from 8:00 a.m. on July 5 to 8:00 a.m. on July 6, 24-hour rainfall totals at Shitang, Taoxu, and Xiaoyi in Hengzhou reached approximately 565.0, 445.0, and 411.5 mm, respectively; Bafeng Village in neighboring Binyang County recorded 713.3 mm. A July 17 expert interpretation published by China Daily further reported an average rainfall of about 749 mm over the Liulan Reservoir catchment and a maximum 24-hour rainfall of 862.5 mm at the Gulou station, and noted that antecedent soil moisture was already near saturation. These data show that the rainfall core was highly localized and concentrated over a short duration. Even when a regional heavy-rainfall threat is foreseeable, it may remain difficult to predict precisely which township, tributary catchment, or reservoir headwater area will receive the most intense rainfall (Source 41).

A preliminary order-of-magnitude estimate based only on early publicly reported point rainfall, a 195 km² drainage area, and an assumed runoff coefficient of 0.75–0.85 yields a 24-hour runoff volume of roughly 60–94 million m³; this is not a formal reservoir-inflow reconstruction. The July 17 expert interpretation reported that professional calculations produced a peak inflow of approximately 4,920 m³/s and a maximum 24-hour flood volume of about 135 million m³. These values were reported as approximately 2.8 times the 1,740 m³/s discharge capacity corresponding to the check-flood condition and 1.8 times the 73.15 million m³ 24-hour flood volume under that condition. This assessment is more complete than the early screening estimate, but it remains

a publicly reported expert interpretation and should ultimately be verified using the rainfall, hydrologic, and reservoir-routing records adopted by the formal investigation (Source 41).

Public interviews reported that the reservoir held about 50 million m³ on the morning of July 5 and was below the flood-season limiting level; another public figure gives an effective storage capacity of about 66 million m³. This indicates that the reservoir was not “full” before the event. The July 17 expert interpretation further stated that the water level at 8:00 a.m. on July 5 was approximately 105.43 m—about 0.97 m below the 106.5 m spillway-crest elevation used in that analysis—so opening the gates at that time would not yet have produced gravity flow over the crest. This explanation still requires confirmation from design drawings, original water-level records, and gate-structure information. Total storage, effective storage, and the flood-control storage actually available at the onset of the storm are not the same concept and cannot be substituted by simply taking a difference between published capacities. A formal reconstruction must use the initial routing level, elevation–storage curve, flood-season limiting level, operating rules, and complete inflow hydrograph. Definitions of relevant water levels and structural elevations are provided in Appendix 2 (Appendix Table 2) (Sources 39 and 41).

2.2 Operational Timeline and Actual Gate Performance

Based on the reservoir manager’s July 13 interview and other public reports, Table 1 summarizes several key points before and during the event. All times, water levels, discharges, and gate states remain subject to confirmation using original duty logs, automatic or manual water-level records, gate-operation records, and the formal investigation findings.

Publicly reported water-level data contain clear inconsistencies. One early account stated that the water level was 111.20 m at 2:30 a.m. on July 6 and was “0.91 m above the design flood level.” A later interview stated that at 6:48 a.m., the water level reached 112.16 m and only then exceeded the design flood level by about 0.05 m. A gauge-room sign photographed by a Jiemian News reporter on July 8 showed a flood-season limiting level of 108.9 m, a design flood level of 112.11 m, and a check flood level of 113.51 m. The reservoir manager later explained that the emergency notice correctly recorded the water level as 111.20 m but that “above” should have read “below.” Thus, 111.20 m was actually 0.91 m below the design flood level; the subsequent on-site sign and the manager’s explanation support this correction (Sources 39 and 40). The reported water level at 10:10 p.m. on July 5 also appears in different circulated versions as either 109.42 m or 105.42 m. The July 17 expert interpretation reported a level of about 105.43 m at 8:00 a.m. on July 5 and a spillway-crest elevation of about 106.5 m, whereas the Southern Weekly account’s 109.42 m at 10:10 p.m. is more consistent with its description that the water had reached or passed the tops of the gates. The 105.42 m figure may therefore reflect a time-point displacement, transcription error, or typesetting error. This brief provisionally uses 109.42 m, but the original water-level record is needed for confirmation. The distinctions among water-level names and structural elevations are explained in Appendix 2 (Sources 39–41).

The manager’s interview indicates that operators planned or were directed to fully open all five gates, but the gates did not all reach the fully open position. Gate No. 5 malfunctioned in the early morning, and Gate No. 3 reportedly slid down and closed automatically at one point. During the full-opening attempt, Gate No. 5 could be raised only about 2 m, compared with an approximate gate height of 2.9 m. The post-event gate positions cannot be used to infer the precise opening history throughout the event. Reporting on the Liulan Reservoir 2026 Flood-Season Operating Plan further indicated that the spillway originally had no gates and that five gates were added during the 2010 rehabilitation. If accurate, the gates and hoisting system that exhibited operating failures were only about 16 years old—far younger than the dam itself. This illustrates that the system component controlling risk need not be the component with the greatest calendar age and provides a concrete example of the “effective age” concept discussed in Section 8.5. The July 17 expert interpretation stated that the project passed acceptance for putting the main works into operation in December 2010 and that the completed works were rated qualified. This does not necessarily conflict with other reporting that final completion acceptance had not been finished, because the two statements may refer to different stages of acceptance; the complete project records are needed for clarification (Sources 39 and 41).

Table 1. Key Timeline of the Liulan Reservoir “7·6” Dam Failure Event (Compiled from Public Reports)

Time	Process Reported in Public Sources
After 3:00 p.m., July 5	Reservoir service-station staff began flood-routing preparations and went on full alert.
About 5:50 p.m.	A flood-routing warning was issued. Releases were originally planned to begin at 10:00 p.m., but the schedule was advanced because of rising water levels and upstream warnings.
About 8:30–8:40 p.m.	Emergency response was activated and staged gate opening began. Reported initial releases were controlled at approximately 50–200 m ³ /s.
About 10:10 p.m.	The reported reservoir level was approximately 109.42 m and continued to rise.
About 2:22 a.m., July 6	Downstream townships were notified to prepare for evacuation under different scenarios. Communications, email, and road conditions were already being affected.
About 5:07 a.m.	After receiving instructions from the higher-level water authority, a warning was issued to fully open all five spillway gates.

Time	Process Reported in Public Sources
About 5:45–5:50 a.m.	The reported level was approximately 111.61 m. During the attempt to fully open the gates, faults were found in some gates or hoisting equipment.
About 6:18 a.m.	Because of severe vibration in the gate structure and increasing safety risk, all control-room personnel were ordered to evacuate. Gate positions could no longer be adjusted thereafter.
About 6:48 a.m.	The reported level was approximately 112.16 m and was said to be about 0.05 m above the design flood level.
About 8:00–8:30 a.m.	Multiple reports and eyewitness accounts described overtopping or clearly observable flow over the dam crest.
About 9:00 a.m.	Reservoir work records were reported to show that an approximately 30 m breach first formed in the eastern section.
About 9:20 a.m.	Media reports cited a manually estimated level of approximately 116.26 m; both the value and elevation datum require formal confirmation.
About 10:20–10:30 a.m.	A larger breach developed in the central section and rapidly enlarged; reported width was approximately 100–110 m.

The reported downstream safe release rate of approximately 220 m³/s helps explain why releases were initially increased in stages: suddenly opening all gates before downstream evacuation was complete could have caused earlier channel overflow and village inundation. Operations therefore faced a genuine tradeoff between controlling reservoir level and avoiding a sudden increase in downstream discharge before evacuation was complete. Whether a more effective pre-release or earlier evacuation strategy was possible can be determined only through complete reservoir-routing and emergency-response reconstruction, not from a few isolated time points. The July 17 expert interpretation also reported that recalculation showed overtopping by more than 0.9 m would still have occurred under the extreme rainfall even if the gates had been opened earlier. This was a public expert assessment rather than the formal investigation finding, but it preliminarily suggests that “earlier gate opening” might not have prevented overtopping itself, although it could have affected overtopping depth and duration and the downstream flood hydrograph—exactly the type of counterfactual question proposed in Section 8.1. The expert interpretation further reported that the reservoir rose by more than 9 m overnight and that the maximum rise during the final hour before breaching was 2.13 m (Source 41).

Public sources still conflict on the service-spillway parameters. Several reports describe it as a five-bay gated ogee spillway excavated through a natural saddle, with an entrance clear width of approximately 60 m.

Some reports give a crest elevation of about 105.70 m; the July 17 expert interpretation used 106.5 m. Other early sources called 108.10 m the spillway-crest elevation, whereas the reservoir manager's interview identified 108.10 m as the flood-season limiting level. This brief does not treat any unverified key elevation or maximum discharge as conclusive. The parameters should be established from design drawings, operating rules, safety-appraisal records, and a common elevation datum (Sources 39 and 41).

2.3 Earthfill-Dam Overtopping, Local Breach Initiation, Breach Enlargement, and Emergency Spillway Facilities

The failure sequence at Liulan Reservoir is broadly consistent with the general mechanics of overtopping erosion of an earthfill dam. When the local reservoir surface reaches or exceeds the crest control elevation, water may pass over the crest and flow down the downstream slope. Once slope protection is damaged, embankment soil is exposed and concentrated gullies may migrate upstream. After erosion cuts into a critical embankment section, local slumping and rapid breach enlargement may occur. This general mechanism, however, cannot replace a site-specific investigation of the initial breach location, local elevations, abutment interfaces, and pre-existing project condition.

Based on public reports, eyewitness accounts, and information attributed to reservoir work records, the most defensible provisional description is as follows: at about 8:00 a.m., the water level began to reach the dam crest and overtopping appeared; concentrated erosion, local collapse, and an initial breach developed at the east abutment almost simultaneously or shortly thereafter; by about 8:30 a.m., floodwater was already discharging through the eastern breach; by about 9:00 a.m., the eastern breach was reportedly about 30 m wide; and between about 10:20 and 10:30 a.m., another, larger breach formed in the central section and rapidly expanded to approximately 100–110 m. Reports differ on the exact times, directions, and dimensions. It cannot yet be strictly demonstrated that overtopping continued for a distinct period before breaching began at the east abutment. A formal reconstruction should use dam-axis stationing, a common elevation datum, time-stamped continuous imagery, drone surveys, and cross-sectional investigations to establish the precise sequence and rate of development.

As shown in Figure 1, pre-event satellite imagery indicates an open bypass-like channel outside the east abutment. Public interviews and rehabilitation-document leads suggest that this feature was likely an emergency spillway whose entrance was closed by a fuse-plug embankment intended to breach at high reservoir levels, with manual blasting or mechanical excavation available if necessary. During this event, the main dam overtopped and breached, but the fuse-plug embankment did not breach as intended and could not be activated manually in time. Online water-conservancy gazetteer materials provide elevations for the fuse-plug crest, activation level, and dam crest that show certain numerical relationships to on-site signs and media estimates, but the source quality, elevation datum, and accuracy have not been reconciled. If all values use the same datum and are accurate, the failure of this facility to activate requires focused explanation in the formal investigation. At this stage, however, the exceedance above the control elevation cannot be quantified directly, and numerical

proximity alone cannot establish the design relationship among these elevations (see the publicly reported values in Appendix Table 3 of Appendix 2).

Media reports citing 2009 rehabilitation information and satellite imagery suggested that timber processing, a village access road, repeated vehicle loading, soil densification, or local filling may have altered the control elevation and erodibility of the fuse plug. No blasting charges had been pre-positioned; manual blasting required approval; communications failed; and excavators could not reach the site because roads were damaged. These facts can help explain why manual activation was not achieved in time, but they do not prove that failure of the fuse plug was the sole cause of the main-dam breach. Design drawings, material testing, field elevation surveys, and counterfactual reservoir-routing calculations are needed to estimate how much timely activation would have reduced the peak reservoir level, delayed overtopping, and potentially changed the final outcome.

As shown by the planform in Figure 1, the reservoir shoreline and dam axis form a degree of planform convergence near the east abutment, where a hillside channel and suspected emergency spillway are also present. This geometry alone does not prove the existence of high-velocity “funnel-like” flow immediately upstream of the dam, because reservoir flow near the dam is normally dominated by storage and relatively low velocities. However, if the east abutment contained a locally low crest or road elevation, a microtopographic depression, an abutment interface, or a downstream flow-guiding channel, overtopping flow could have concentrated there first. Once an initial gully or local collapse formed, a positive feedback could have developed: lower elevation → greater flow concentration → accelerated erosion → breach enlargement. This remains a testable hypothesis and should be evaluated using detailed topographic surveys, continuous imagery, and coupled two- and three-dimensional hydraulic and erosion analysis.

2.4 Sediment Transport and Secondary Water-Quality Impacts

Reservoir overtopping and dam breaching release not only floodwater but potentially large quantities of watershed erosion sediment, reservoir deposits, and embankment soil. The yellow-brown floodwater in Figure 2 and thick mud deposits visible in other post-event imagery indicate substantial sediment deposition and erosion in downstream communities, farmland, roads, and channels.

Sediment-laden flooding can alter channel cross sections, block culverts and drainage systems, scour bridge foundations and roadbeds, and affect drinking-water sources, wells, agricultural topsoil, and aquatic ecosystems. Public reports also describe rescue boats threatened by submerged trees, broken poles, and building debris. Recovery should therefore treat sediment removal, classification and disposal of deposits, water-quality testing, bridge and culvert inspection, bank stabilization, and public river-recreation safety as priorities equal to treatment of the dam itself.



Figure 1. Pre-Event Layout of Liulan Reservoir and Satellite Image of the Channel Near the East Abutment. The open bypass-like channel outside the abutment is visible on the right. Public reporting has associated it with a fuse-plug emergency spillway, but its formal function, control elevation, and relationship to the initial breach remain to be confirmed from design records and the investigation. Image source: Google Maps (imagery credits shown in the figure).



Figure 2. Aerial View of the Dam Breach and Near-Dam Downstream Area after the Liulan Reservoir Failure. The image shows the large breach, sediment-laden outflow, and affected near-dam downstream area. It documents the overall post-event scene but cannot by itself establish the initial breach location, the precise sequence of overtopping and local breaching, peak discharge, or the specific failure mechanism. Image source: Red Star News, “The Critical 12 Hours before and after the Liulan Reservoir Breach: Those Trapped and Their Rescuers,” July 11, 2026.

3. Basin Relationships and Attribution of Multiple Flood Sources

3.1 Combined Flooding from Liulan, Yunbiao, and the Wider Region

Yunbiao Reservoir lies approximately 5 km west of Liulan Reservoir. On July 6, both reservoirs overtopped and developed breaches, and their downstream channels converged near Naguang Village before entering the Yu River. Other small and medium reservoirs in the region also experienced emergencies during the same period. Public reports indicate that the inundation affecting Xiaoyi, Yunbiao, Maling, and areas in the direction of Guigang was jointly influenced by floodwaters from the mountainous upper reaches of the Zhenlong River, tributary inflows, breach outflows from the two reservoirs, and elevated water levels in the regional river system.

Accordingly, without models and field measurements, the inundation depth, arrival time, and casualties at any particular downstream village or town cannot all be attributed to the single Liulan breach. Formal attribution should use one- and two-dimensional models with multiple inflow boundaries and should be calibrated against high-water marks, videos, remote-sensing inundation extents, and flood-arrival times.

3.2 Limits of Inference Regarding the Datengxia Project, the Pinglu Canal, and the May Earthquake

Floodwater from Liulan Reservoir enters the Yu River through the Zhenlong River and subsequently joins the Xi River system. The Datengxia Water Conservancy Project is located farther downstream on the Qian River. Based on presently available spatial information, elevation relationships, and public records, there is no evidence that Datengxia backwater extended to the area below Liulan Dam, and still less evidence that it contributed causally to overtopping of the reservoir.

The northern terminus of the Pinglu Canal begins at the Pingtang River mouth in the Xijin Reservoir area of Hengzhou, connecting with the Yu–Xi River system. The canal then crosses a drainage divide and reaches the Beibu Gulf through the Jiuzhou and Qin rivers; it is not a direct flood-conveyance route for Liulan Reservoir. Any assessment of whether trial water conveyance or operation of related control structures affected regional water levels or flows during the event must be based on contemporaneous operating records and regional hydraulic modeling, not on spatial proximity alone.

Post-earthquake inspections and pre-flood-season checks at Liulan Reservoir would have been appropriate following the May 2026 Guangxi earthquake. However, publicly available information has not shown that the earthquake caused abnormal settlement, cracking, seepage, pore-pressure changes, or damage to spillway facilities, and no evidence currently establishes a direct causal link between the earthquake and the July 6 breach. The formal investigation should still review site ground-motion estimates, post-earthquake inspection records, and any anomalies in embankment deformation, seepage, abutment interfaces, or gate-hoisting systems. On the evidence now available, however, the more clearly established direct sequence was extreme rainfall, rapidly

increasing inflow, a continuing rise in reservoir level and overtopping, followed by erosion of the embankment and rapid initiation and enlargement of local breaches.

4. Water-Supply Source, Agricultural Production, and Basin Management

Liulan Reservoir is both an irrigation and regional water-supply project and is closely linked to Hengzhou's jasmine industry and rural livelihoods. Public reports show that extensive jasmine fields, agricultural-supply shops, roads, village water-supply facilities, and homes in Yunbiao, Xiaoyi, and nearby areas were inundated. After the flood recedes, agricultural recovery must include not only drainage and replanting but also testing for soil salinity and contaminants, disease control, restoration of agricultural topsoil, and reconstruction of agricultural-input and processing chains.

Pre-existing soil erosion, nonpoint-source pollution, and drinking-water-source protection issues in the reservoir and upstream catchment are not part of the same direct causal chain as the overtopping event, but they can influence sediment yield, water quality, and recovery cost. Future management should integrate hillslope soil and water conservation, riparian buffers along inflowing streams, reservoir-bank stability, drinking-water protection, and extreme-flood safety.

5. Post-Disaster Recovery: From Emergency Response to Functional Reconstruction

5.1 Safety of the Dam and Infrastructure

The first post-disaster task is a safety assessment of the remaining embankment, breach side slopes, foundation, through-dam or adjacent structures, spillway, and gate systems. Based on topographic surveys, geologic investigation, seepage assessment, and stability analysis, temporary diversion, breach protection, and measures to prevent secondary collapse should be implemented. Rapid reservoir drawdown may also trigger reservoir-bank landslides, requiring parallel inspection of banks, roads, and structures near the reservoir.

Downstream roads, bridges, culverts, levees, power, communications, water-supply, and gas facilities should not be returned to service on visual appearance alone after scouring and inundation. Reports describe undermined roadbeds, damaged bridges and culverts, communications outages, and flooded water-supply facilities. Service should be restored through structural inspection, flushing and disinfection, electrical testing, and staged recommissioning.

5.2 Sediment Removal, Ecology, and Public Health

Sediment removal should distinguish among flood-conveyance sections of channels, village and town roads, homes, farmland, and potentially contaminated deposits. Drinking-water sources, temporary intake points, wells, and inundated water-supply facilities should be tested for microorganisms, turbidity, organic

matter, and necessary chemical indicators. Submerged debris, deep holes, concealed channels, and unstable banks should be inspected and marked before water-contact recreation resumes. A concise downstream river-safety advisory is provided in Appendix 3.

5.3 Recovery of People, Industries, and Communities

The State Council announcement of July 24 stated that the dam failure disaster caused major casualties. According to CCTV News, The Beijing News, and other reports, the disaster statistics released by Nanning on July 9 stated that the Liulan Reservoir breach flood alone had caused 26 deaths and left 7 people missing, while the corresponding regional totals were 39 deaths and 9 missing. This was the first official separate statistical count for the Liulan Reservoir breach flood, but it may still change as identities are confirmed, missing persons are located, and disaster attribution is refined. Dynamic statistics for the broader regional rainstorm disaster should not be treated as the Liulan Reservoir event's standalone attributable toll. The final figures for deaths, missing persons, and losses should follow the State Council investigation and final official verification (Source 3).

Public reporting clearly shows that the consequences extended far beyond the dam itself, affecting housing, farmland, the jasmine industry, agricultural supplies and processing businesses, transportation, schools, water supply, and community confidence. Recovery should address resettlement, compensation, housing reconstruction, agricultural production, psychological support, risk communication, and transparent public engineering assessment.

6. Monitoring, Forecasting, Operations, Warning, and Evacuation Systems

Public interviews indicate that the problem was not simply that “the rainfall forecast was insufficiently accurate.” The event also exposed fragility in the data and communications chains: upstream hydrologic-station data stopped updating during the critical period; reservoir staff relied mainly on manual staff gauges and experience-based inflow estimates; automated monitoring and rainfall-reporting systems had experienced failures or transmission problems; power outages, network disruptions, road closures, and even email-storage limitations affected information receipt; and the reservoir station, township governments, villages, and evacuation operations lacked a confirmation loop that could remain functional under extreme conditions.

The “four forecasts” system—forecast, early warning, simulation, and contingency planning—should be upgraded from isolated equipment to an end-to-end operational chain: radar and numerical weather prediction → catchment-average rainfall → runoff generation, concentration, and reservoir inflow → reservoir routing → actual gate openings and outflow → downstream inundation prediction → confirmation that warnings were received → evacuation and rescue. Critical equipment should have backup power, offline recording, satellite or other emergency communications, and training and exercises for manual gauge reading, manual gate operation, and loss-of-power conditions.

Annual gate operation does not, by itself, demonstrate that reliability under extreme operating conditions has been verified. For high-consequence reservoirs, full-stroke functional tests, brake and self-locking inspections, hoisting-force tests, verification of backup power and manual operation, and other appropriate checks should be conducted periodically under safe conditions, with traceable records of gate opening versus time maintained. Emergency spillway facilities should likewise be periodically inspected to verify their control elevations, downstream conveyance paths, potential roadway obstructions, soil compaction, and conditions required for manual activation. The fact that such facilities have “not been used for decades” should not be regarded as evidence of their reliability.

Operating plans should define actions at specific thresholds: when to pre-release, when to lower the flood-season limiting level, when to begin early evacuation, when to enter dam-preservation mode, when to activate emergency spillways, and who has decision authority and how authorization and execution are confirmed when communications and power fail. Optimal decisions during extreme floods are rarely the responsibility of a single discipline; they require joint judgment by hydrologists, hydraulic engineers, geotechnical and mechanical specialists, operators, emergency managers, and local government.

7. Recovery Options Based on Functional Needs and Risk Tradeoffs

Liulan Reservoir’s irrigation and water-supply functions are highly important, but restoring those functions need not mean rebuilding the original project in the same form. Repair in place, upgrading standards, lowering the operating level, reclassification and restricted use, partial retirement, or construction of alternative water sources should be considered only after the cause of the disaster is established, downstream risk is updated, and life-cycle costs are compared. The principal consequences of lost functions, substitution options, and urgency of recovery are summarized in Table 2.

If repair in place is selected, the following questions require focused analysis: whether the design and check floods should be updated; whether service and emergency spillway capacities are adequate; whether gates should be replaced and redundant power added; whether crest and downstream-slope overtopping protection should be installed; whether the crest elevation and wave freeboard should be modified; whether seepage-control and drainage improvements are needed at the embankment, foundation, and structure interfaces; and how downstream flood-risk zones and evacuation routes should be redelineated.

Table 2. Principal Functions of Liulan Reservoir, Substitution Options, and Recovery Urgency

Function	Principal Consequences if Not Restored	Potential Substitutes	Recovery Urgency
Irrigation	Water supply for approximately 157,300 mu of farmland would be affected, and benefits of existing irrigation-district improvements could not be fully realized.	Regional water transfers, pumping stations, ponds, and temporary sources may provide partial substitution, but complete short-term replacement would be difficult.	High
Water supply	Water-supply reliability and emergency-source capacity for nearby towns and villages would decline.	Alternative sources would require simultaneous construction of pipelines, pumping stations, and water-quality safeguards.	High
Flood control	Existing storage and regulation capacity would decline, while the safety margin of the previous system has also been called into question.	Alternatives should compare new storage, channel improvements, warning and evacuation, and downstream engineering measures.	Requires reassessment
Hydropower	Small-hydropower revenue and equipment would be lost or damaged.	The regional power system can provide substitution.	Relatively low
Ecology, fisheries, and related industries	Aquatic ecology, fisheries, jasmine production, and related livelihoods would be affected.	Recovery can proceed gradually with water-source, farmland, and ecological restoration.	Medium to high

8. Recommendations for Investigation, Research, and Engineering Reassessment

8.1 Reconstruction of the Hydrologic–Operational–Discharge Process

The formal investigation should establish a unified timeline and reconstruct catchment-average rainfall, upstream inflows, the reservoir-inflow hydrograph, reservoir levels and storage, actual opening of each gate, discharges through service and emergency spillways, breach initiation, and downstream water levels. It should distinguish among nominal design discharge, theoretical available discharge under the actual head and tailwater conditions, and actual discharge constrained by equipment malfunction, blockage, operations, and downstream limitations.

At least four counterfactual scenarios should be compared: actual operation; all gates operating normally according to the original plan; earlier pre-release combined with earlier completion of downstream evacuation; and activation of the fuse-plug emergency spillway as designed. For each scenario, the maximum reservoir level, onset and duration of overtopping, and downstream flood consequences should be calculated. Only then can the question be answered whether earlier gate opening or fuse-plug activation would have changed the outcome.

Once complete data are available, the actual sustained outflow and unit discharge per effective width, $q = Q/B$, can be calculated to evaluate hydraulic performance of the spillway facilities. This metric must be based on a verified total outflow hydrograph, effective flow width, gate openings, hydraulic head, and tailwater conditions; it cannot be obtained by simply dividing a nominal maximum design discharge.

8.2 Two-Dimensional Reservoir and Local Three-Dimensional Hydraulic Modeling of Overtopping-Flow Concentration

After reconstructing the overall water balance and reservoir routing, a two-dimensional hydraulic model of the reservoir should be developed using bathymetry, shoreline geometry, detailed near-dam elevations, crest microtopography, service and emergency spillway layouts, and inflow boundary conditions. The model should simulate flood propagation through the reservoir, the near-dam planform velocity field, local water-surface gradients, discharge allocation among outlets, and whether planform convergence at the east abutment could have combined with local low points to concentrate overtopping flow. One-dimensional routing or level-pool analysis is suitable for reconstructing the overall reservoir balance but cannot by itself resolve local flow direction and velocity distribution.

For the east abutment, service spillway, emergency spillway, abutment–embankment interfaces, and area of the initial breach, local three-dimensional hydraulic models should be considered to evaluate vertical circulation, local acceleration, recirculation, turbulence, wind waves or wind setup, crest overflow, and downstream-slope flow. Where necessary, these models should be coupled with soil erosion, slope-protection failure, and breach-enlargement models. Calibration should use measured water levels, gate openings, wind fields, videos, drone imagery, post-event topography, and high-water marks. Sensitivity and uncertainty analyses should address bathymetry, local elevations, roughness, and erosion parameters. Modeling should test mechanistic hypotheses, not replace field investigation.

8.3 Breach-Flood and Multiple-Source Inundation Modeling

A one- and two-dimensional hydraulic model should be developed for the Zhenlong River–Yu River corridor, including inflow boundaries from Liulan, Yunbiao, and other important sources. Key inputs should be disclosed or recorded: breach location, final width, invert elevation, development time, outflow hydrograph, digital elevation model accuracy, roughness, downstream boundary conditions, and bridge and culvert backwater effects.

The model must be calibrated against field high-water marks, remote-sensing inundation extent, flood-arrival times, media videos, drone imagery, and measured water levels. Sensitivity and uncertainty analyses should be conducted for breach parameters, roughness, and boundary conditions lacking field measurements. Uncalibrated animations may support communication, but they should not be presented as the actual flood process.

8.4 Design Standards, PMP/PMF, and Nonstationary Reassessment

The design flood, check flood, and extraordinary-operation flood standards used in the original design, rehabilitation projects, and operating plans should be reviewed. Based on applicable regulations, project scale, and downstream consequences, the investigation should determine whether probable maximum precipitation/probable maximum flood (PMP/PMF) analysis or other extreme-flood scenarios are required. Available information is insufficient to determine whether this event approached PMP, and “a one-in-a-thousand-year event” should not be treated as a confirmed conclusion. One local line of evidence for reassessment is the expert report that a surveyed flood mark at Zhenlong Hydrologic Station, approximately 15 km upstream, reached 139.81 m—7.05 m above the previously surveyed historical maximum flood level from 1935 (Source 41).

Under nonstationary climate conditions, design standards require adaptive reassessment, but higher standards cannot eliminate all risk. The first line of defense is to update the design flood, discharge capacity, crest freeboard, and overtopping protection. The second is to manage residual risk once standards are exceeded, including failure modes, warning and evacuation, emergency releases, cascading infrastructure impacts, and post-disaster functional recovery.

8.5 The “Effective Age” of the Dam System and Risk Prioritization

Calendar age is not the only measure of dam risk. Embankment materials, foundations, spillways, gates, monitoring systems, the hydrologic design basis, maintenance, and management arrangements may be viewed as multiple “clocks” running in parallel. Whichever component first exhausts its safety margin may become the controlling factor. This “effective age” concept is useful for risk communication and asset management, but it is not a single numerical indicator that can replace formal safety appraisal.

Regional risk prioritization should consider dam height, reservoir storage capacity, downstream population and assets, dam type, flood design standards, effective discharge capacity, gate reliability, monitoring and communications, historical deficiencies, land-use change, and management capacity. Rehabilitation, inspections, full-stroke gate testing, updated flood studies, and emergency exercises are, in effect, measures to restore the system’s remaining safety margin. A long history of stable operation should not substitute for an assessment of current conditions.

8.6 Comparison of Ecological, Public-Health, and Recovery Options

Recovery alternatives should be compared using multiple objectives, including life-cycle cost, risk-reduction benefit, irrigation and water-supply reliability, ecological impact, social acceptance, and operation and maintenance capacity. The assessment should also address downstream sediment, drinking-water safety, sand and silt deposition on farmland, recovery of jasmine and other industries, vector-borne disease, and mental health.

9. Conclusions and Broader Lessons: Design Standards, Residual Risk, and Collective Engineering Judgment

The Liulan Reservoir event demonstrates that extreme hydrologic loading, the overtopping vulnerability of earthfill dams, the actual reliability of spillway facilities, reservoir operations, and warning and evacuation are not independent of one another. The probability chain leading to severe consequences can be expressed conceptually as the probability of a flood exceeding the engineering design standard, multiplied by the conditional probability of overtopping under that flood, multiplied by the conditional probability that overtopping initiates and enlarges a breach, and finally multiplied by the conditional probability that people, assets, and infrastructure within the breach-flood impact area are exposed. Different engineering and management measures act on different links in this chain. To quantify risk more fully, this probability chain must be combined with the consequences of exposure, including casualties, economic losses, infrastructure disruption, and broader social impacts.

A “100-year event” does not mean that a project will necessarily remain safe for 100 years. Under the simplified assumption of an independent 1% annual exceedance probability, the probability of at least one event exceeding that standard in 50 years is approximately $1 - (0.99)^{50} \approx 39.5\%$. A design standard represents an engineering balance between risk and cost; it is not a guarantee of absolute safety.

The sector should advance two efforts simultaneously. First, engineering protection standards should be reassessed using updated hydrologic, climate, land-use, and downstream-exposure information. Second, regardless of whether standards are upgraded, residual risk beyond the standard must be managed. Relevant compound scenarios include gate or power failure, nonactivation of emergency spillway facilities, communications failure, road blockage, refusal to evacuate, and simultaneous emergencies at multiple reservoirs.

Dam safety is inherently multidisciplinary and multi-institutional. It requires participation by hydrologists, hydraulic engineers, geotechnical and structural engineers, mechanical and electrical specialists, reservoir operators, emergency managers, transportation and communications agencies, public-health professionals, and local communities. The formal investigation should avoid explaining the entire event from a single disciplinary perspective or by a single failure. Likewise, decades without an incident cannot be treated as proof that adequate safety margin remains.

The State Council investigation and assessment team called for a “comprehensive reconstruction of the disaster process, a full and factual determination of the causes, and an end-to-end examination of problems and responsibilities in disaster prevention, response, and emergency management.” The systems-risk framework used in this brief is consistent with that mandate. All provisional interpretations should ultimately be corrected or confirmed by the formal investigation findings (Source 9).

Sources, Methods, and Scope

This brief was prepared using official announcements, press-conference information, meteorological and hydrologic data, publicly available project-document leads, news interviews, satellite and aerial imagery, and professional discussions that could be obtained and cross-checked through July 25, 2026. The list below includes materials actually used or consulted for cross-checking. Where the same official announcement was republished by multiple outlets, preference is given to the official, original, or most complete version. News reports and professional discussions can provide valuable factual leads and research questions, but they cannot substitute for original hydrologic records, design documents, gate-operation records, field investigation, or the State Council’s formal investigation findings.

I. Official Announcements, Meteorological and Hydrologic Information, and Investigation Materials

1. Nanning Municipal Emergency Management Committee: July 6, 2026 briefing on flood-control conditions in Nanning, overtopping and breaches at Liulan and Yunbiao reservoirs, overtopping at Liuwang Reservoir in Binyang County, and escalation of the flood-control emergency response from Level III to Level I.
2. Nanning Municipal Emergency Management Committee, “Notice Declaring Certain Flood-Affected Areas to Be in an Emergency Flood-Control Period,” July 6, 2026.
3. Public briefings from the first, second, and third Nanning flood-control and disaster-relief press conferences, July 6, July 7, and July 9, 2026.
4. Public briefing from the Guangxi Zhuang Autonomous Region emergency rescue and disaster-relief press conference, July 7, 2026.
5. Guangxi Meteorological Observatory and Nanning Meteorological Observatory: observed rainfall, forecasts, and historical extremes for Guangxi and Nanning during Typhoon “Maysak,” July 3–7, 2026.
6. Guangxi Hydrological Center: public information on water conditions, stations above warning level, and flood warnings for the Yu River and other major rivers in Guangxi, July 5–8, 2026.
7. Guangxi Department of Water Resources, Ministry of Water Resources, and State Flood Control and Drought Relief Headquarters: public information on emergency flood-response levels, reservoir operations, and rescue activities during Typhoon “Maysak,” July 6–8, 2026.
8. Hengzhou Liulan Reservoir Service Station: emergency notices concerning full opening of the spillway gates and related operating information, republished or shown by multiple media outlets, July 5–6, 2026.

9. Ministry of Emergency Management / Chinese Government website, “State Council Establishes Investigation and Assessment Team for the Liulan Reservoir ‘7·6’ Dam Failure Disaster in Hengzhou, Nanning, Guangxi,” July 24, 2026.

II. Project Information, Project Records, and Spatial Imagery

10. Local water-conservancy gazetteers and online engineering materials concerning Liulan Reservoir: construction dates, drainage area, dam type, dam height, crest elevation, spillway configuration, and nominal maximum discharge. The original editions and some parameters have not been independently verified; this brief treats them as subject to confirmation.

11. Public tender, design, and project-reporting materials for the Liulan Reservoir rehabilitation around 2009, together with publicly available information concerning the 2020 safety appraisal and subsequent repairs.

12. Publicly available information and media summaries on the continued upgrading and modernization of the Liulan Reservoir Irrigation District, including irrigated area, investment, and reservoir functions.

13. Google Maps satellite imagery, Imagery ©2026 CNES / Airbus, Maxar Technologies; used to identify the main dam, service spillway, open channel near the east abutment, and downstream spatial relationships.

III. Initial Reporting and Field Information, July 6–8, 2026

14. Xinhua News Agency, “Serious Emergency at Liulan Reservoir in Hengzhou, Nanning, Guangxi,” July 6, 2026.

15. Xinhua News Agency, “Multiple Reservoir Emergencies in Nanning, Guangxi; Local Authorities Organize Evacuations,” July 6, 2026.

16. Xinhua News Agency, “Hengzhou, Guangxi: Breach Develops in Reservoir Dam as Nearby Residents Are Urgently Evacuated,” July 6, 2026.

17. Xinhua News Agency, “Feature: Rescue in Liulan Village,” July 7, 2026.

18. CCTV.com, “Nanning, Guangxi Reports Flood Conditions: Overtopping and Breaches at Liulan and Yunbiao Reservoirs in Hengzhou; Overtopping at Liuwang Reservoir in Binyang,” July 6, 2026.

19. CCTV News app, “Multiple Reservoir Emergencies in Hengzhou, Guangxi; Flood-Control Response Raised to Level I,” July 6, 2026.

20. CCTV News app, “Drone View of Liulan Reservoir: Homes Inundated by Floodwater,” July 6, 2026.

21. CCTV News, “CCTV Reporter Visits the Core Emergency Area at Liulan Reservoir,” July 7, 2026.

22. The Paper, “Overtopping and Breaches at Liulan and Yunbiao Reservoirs in Hengzhou, Nanning; Overtopping at Liuwang Reservoir in Binyang,” July 6, 2026.

23. The Paper, “On Scene | Liulan Reservoir Breach Stabilized, Floodwater Receding, Residents Clear Mud and Rebuild,” July 7, 2026.

24. Yicai, “Reservoir Emergency in Hengzhou, Guangxi; DJI Drones Used to Evacuate Trapped Residents,” July 6, 2026.

25. Time Weekly, “Liulan Reservoir Dam Failure in Guangxi: Irrigation-District Modernization Began Three Years Earlier and Diversion Dam Was Reinforced,” July 6, 2026.
26. Qilu Evening News / Qilu One Point, “Video | Breach Develops in Liulan Reservoir Dam; Villager Reports Being Trapped at Home without Water or Electricity,” July 6, 2026.
27. Qilu Evening News / Qilu One Point, “Investigation | Reviewing the Liulan Reservoir Dam Failure: A Sixty-Six-Year-Old Earth Dam Meets Record Extreme Rainfall,” July 7, 2026.
28. Jimu News, “Breach Develops in Liulan Reservoir Dam in Hengzhou, Guangxi; Villagers Gather Supplies for Rescue Efforts,” July 6, 2026.
29. Jimu News, “Field Visit to Liulan Reservoir: Water Still Discharging through the Dam Breach; Villagers Spent the Night on a Mountain,” July 7, 2026.
30. Red Star News, “Liulan Reservoir Staff Recall the Moment of Failure: Water Far Above Design Level, All Staff on Duty and Gates Opened Early—‘It Was Rising Every Minute,’” July 7, 2026.
31. Red Star News, “Yabei Village Downstream of the Failed Reservoir: No Power or Network; Some Villagers Spent the Night on a Mountaintop as Hengzhou Organized Staged Evacuations,” July 7, 2026.
32. China Newsweek, “Liulan Reservoir Had Once Been Officially Called a ‘Problem Reservoir’; Villagers Say It Was Full the Day before the Emergency,” July 7, 2026. Residents’ recollections and the phrase “problem reservoir” are treated only as investigative leads, not as an official technical classification.
33. The Beijing News, “Liulan Reservoir Emergency in Two Minutes: Collapse Continued on Both Sides of the Breach; Rehabilitation Project Had Required Design for a 100-Year Flood,” July 7, 2026.
34. Southern Weekly, “Floodwater Submerges the ‘World Capital of Jasmine’: ‘We Had Never Seen the Entire Village Under Water,’” July 7, 2026.
35. Southern Window / Yan Finance, “Guangxi Floods Undermine the Supply Base of Luckin Coffee and Mixue,” July 8, 2026.
36. Tang Bohu, Phoenix News, “After Overtopping: An Exclusive In-Depth Review of the Liulan Reservoir Dam Failure in Hengzhou, Guangxi,” July 6, 2026. Order-of-magnitude estimates and preliminary mechanistic interpretations in the article are used only as early analytical references.

IV. Subsequent In-Depth Reporting, Interviews, and Analysis

37. Sanlian Lifeweek, “Thirty-Nine Dead in Guangxi Rainstorms: Lives Rapidly Overwhelmed by Flooding beyond Experience,” July 9, 2026. The figure 39 in the title refers to the regional rainstorm-disaster total and is not treated as the standalone toll from the Liulan Reservoir event.
38. Red Star News, “The Critical 12 Hours before and after the Liulan Reservoir Breach: Those Trapped and Their Rescuers,” July 11, 2026. Figure 2 in this brief is also from this report.
39. Southern Weekly, Zheng Dan and Su Luojie, “Liulan Reservoir Manager Describes the 24 Hours before the Breach,” July 14, 2026 (online edition; cited from a PDF copy retained by the author).

40. Jiemian News, “Seven Days after the Liulan Reservoir Overtopping: Operations, Evacuation, and Rescue,” July 14, 2026. Includes the reporter’s July 8 field record of the gauge-room markings at the dam crest: flood-season limiting level 108.9 m, design flood level 112.11 m, and check flood level 113.51 m.
41. China Daily website, “Water-Conservancy Expert Explains the Causes of Overtopping and Breaching at Liulan Reservoir in Hengzhou, Nanning,” July 17, 2026. The report cites hydrologic reassessment results by Xie Hailin, a professor-level senior engineer at the Guangxi Water and Power Design Institute, including catchment-average rainfall, peak inflow, 24-hour flood volume, discharge capacity under the check-flood condition, and an earlier-gate-opening scenario. This brief cites it as a public expert interpretation, subject to final confirmation by the formal investigation.
42. “Technical Review of the Liulan Reservoir ‘7·6’ Overtopping and Dam Failure—When a 66-Year-Old Homogeneous Earthfill Dam Met a ‘One-in-a-Thousand-Year’ Extreme Rainstorm,” July 2026. The article states that it was compiled using AI tools. Its “one-in-a-thousand-year” characterization, estimated discharges, and hydrographs are not used as factual evidence here because they are not supported by original records.

V. Professional Discussion and Methodological References

43. Professional discussions in water resources, hydrology, environmental engineering, and related fields following the original July 8 WeChat post, addressing PMP/PMF, reassessment of design standards, two-dimensional inundation modeling, model calibration, and risk management. These comments were used only to frame research questions and refine the analytical framework, not as factual evidence about the event.
44. Professional discussions surrounding the original LinkedIn post after July 8, 2026 addressed such topics as the “effective age” of dam systems, remaining safety margins, full-stroke functional verification of gates, multidisciplinary assessment, and discharge per unit effective width. The related comments are included only as references for concepts and research methodology.

Note: The list above is limited to materials actually used or consulted for cross-checking. Full-text republications of the same Xinhua, CCTV, or official notices on other platforms are not listed separately. When the formal investigation report, design documents, and original monitoring records become available, they should supersede provisional figures and secondary reporting.

Statement on Artificial-Intelligence Assistance: The author conceived and wrote this brief and is responsible for all content. Artificial-intelligence tools were used to assist with organizing attachments, locating and cross-checking public information, comparing reports, checking numerical consistency, and polishing language. All technical judgments, source selections, and final wording were reviewed by the author, who retains full responsibility for the document.

Appendix 1. Key Data for Liulan Reservoir (Compiled from Public Sources)

Appendix Table 1. Key Data for Liulan Reservoir (Public Information Available through July 25, 2026)

Item	Publicly Reported Data or Description
Location	Liulan Village, Xiaoyi Town, Hengzhou, Nanning, Guangxi
River system	Zhenlong River, a tributary of the Yu River in the Xi River system of the Pearl River Basin
Drainage area	Approximately 195 km ²
Total storage	Approximately 93.19 million m ³ ; another figure is approximately 95.52 million m ³ , pending verification from formal documents
Effective storage	Public reports give approximately 66 million m ³ ; this is not the same as flood-control storage
Dam height and type	Maximum height approximately 42 m; homogeneous earthfill dam
Construction	Started in 1958 and completed in 1960
Service spillway	Public reports describe a five-bay gated ogee spillway with an entrance clear width of approximately 60 m. Key elevations and discharge capacity require confirmation from design documents. Gates were added during the 2010 rehabilitation.
Emergency spillway	Fuse-plug embankment at the entrance to an emergency spillway on the east side; it did not breach as intended during this event, and the cause remains under investigation
Principal functions	Irrigation, water supply, flood control, hydropower, ecology, and fisheries
Irrigated area	Approximately 157,300 mu
Catchment-average rainfall (expert interpretation)	Approximately 749 mm; maximum 24-hour rainfall of 862.5 mm at Gulou station (Source 41; pending formal investigation confirmation)
Peak reservoir inflow (expert calculation)	Approximately 4,920 m ³ /s (Source 41; pending formal investigation confirmation)
Maximum 24-hour flood volume (expert calculation)	Approximately 135 million m ³ (Source 41; pending formal investigation confirmation)

Item	Publicly Reported Data or Description
Discharge capacity under check-flood condition (expert interpretation)	Approximately 1,740 m ³ /s (Source 41; pending formal investigation confirmation)
Official event designation	State Council announcement of July 24, 2026: Liulan Reservoir “7·6” dam failure disaster

Appendix 2. Terminology for Principal Reservoir Water Levels and Structural Elevations

To avoid the confusion of different water-level and structural-elevation terms in public reporting, Appendix Table 2 defines the principal terms used in this brief. These terms represent different operating, design, or geometric concepts and are not interchangeable. Their precise definitions and values should follow applicable standards, design documents, operating rules, and the formal investigation record.

Appendix Table 2. Principal Reservoir Water-Level and Structural-Elevation Terms

Term	General Meaning and Engineering Function	Use and Cautions in This Brief
Flood-season limiting level (flood-control limiting level)	The upper operating level for conservation storage during the flood season, used to preserve flood-control storage and define the starting condition for flood operations.	It is not an absolute safety limit that the reservoir may never exceed. The July 13 interview called 108.10 m the flood-season limiting level; this must be distinguished from the spillway-crest elevation.
Normal pool level	The operating level used under normal conditions to satisfy irrigation, water-supply, hydropower, and other conservation purposes.	Its relationship to the flood-season limiting level depends on season and operating rules; during the flood season, the limiting level is often below the normal pool level.
Design flood level	The maximum reservoir level reached when the design flood is routed according to prescribed operating procedures.	It is not the level at which the spillway first begins to discharge, nor does reaching it imply that failure is inevitable. Published values for Liulan Reservoir still require confirmation using a common datum.

Term	General Meaning and Engineering Function	Use and Cautions in This Brief
Check flood level	The maximum reservoir level expected under the check flood, used as an important control level for dam-safety verification under extraordinary operating conditions.	The on-site sign showed 113.51 m, but the value still requires confirmation from design documents and a common datum. It is not the measured peak level during this event.
Flood-control high water level	The maximum reservoir level reached when a reservoir performs downstream flood-control operations for the design flood of the protected area.	It applies only where the reservoir has the corresponding downstream flood-control function and should not be equated directly with the design flood level.
Spillway-crest elevation	The structural elevation of the spillway control section; it is a geometric parameter, not an operating water-level name.	For a gated spillway, the crest may be lower than the flood-season limiting or normal pool level. Whether discharge occurs and its magnitude also depend on gate opening, head, and tailwater.
Emergency-spillway control elevation	The control elevation at which an emergency spillway, fuse plug, or other auxiliary discharge facility is intended to activate.	The design elevation, activation condition, and reason for nonactivation of Liulan Reservoir's east-side fuse-plug facility should be established from design documents and field investigation.
Dam-crest elevation	The structural elevation of the top of the dam. Local crest elevation may be affected by construction tolerances, settlement, road modifications, and differential deformation.	Overtopping may begin when the local water surface—where relevant including wind setup and wave runup—exceeds the local crest elevation. The lowest crest point, rather than only the nominal elevation, is important.

Using the terminology above, Appendix Table 3 compiles publicly reported values for the main elevations and water levels at Liulan Reservoir available through July 25, 2026. The sources vary in reliability and may not use a common datum. The values are presented only for cross-comparison and remain subject to design documents, operating rules, and the formal investigation.

Appendix Table 3. Publicly Reported Principal Elevations and Water Levels at Liulan Reservoir (Pending Formal Verification)

Elevation/Water-Level Item	Publicly Reported Value (m)	Source and Explanation (All Pending Formal Verification)
Spillway-crest elevation	Approximately 105.70; alternatively 106.5	Some reports and the manager's interview use approximately 105.70 m; the July 17 expert interpretation uses 106.5 m. Both require verification from design drawings and a common datum.
Flood-season limiting level	108.9 (alternative report: 108.10)	A Jiemian News reporter photographed 108.9 m on the dam-crest gauge-room sign on July 8; the manager's interview cited 108.10 m. Both remain in circulation.
Design flood level	112.11	Shown on the gauge-room sign; consistent with the manager's statement that the 6:48 a.m. level of 112.16 m was about 0.05 m above the design flood level.
Check flood level	113.51	Shown on the dam-crest gauge-room sign; not the measured peak level of this event and still subject to confirmation from design documents and a common datum.
Emergency spillway (fuse plug)	Crest approximately 113.5; activation approximately 113.17	Online water-conservancy gazetteer material; source and datum require verification. If these values share a datum with the on-site signs and event levels, the numerical relationship warrants investigation, but it cannot alone establish design correspondence or quantify exceedance depth.
Dam-crest elevation	114.38	Red Star News on July 11, citing a water-conservancy gazetteer; maximum dam height 42 m and foundation described as thick-bedded gray-green argillaceous shale.
Event peak water level	Approximately 116 (manual estimate: 116.26)	Media citation of a manual estimate; both the value and elevation datum require formal verification.

Note: Appendix Table 3 compiles public values from sources of differing reliability and should not be interpreted as a verified "elevation ladder." Published spillway-crest elevations include approximately 105.70 m and 106.5 m; reported flood-season limiting levels include 108.9 m and 108.10 m; and the fuse-plug and peak-event levels have not been verified from original records. Only after a common datum is confirmed and

the values are checked against design documents, surveys, and monitoring records can they be used to determine relative elevations, activation conditions, and overtopping depth.

Special Note: Public sources conflict on whether 108.10 m represents an operating water level or a structural elevation. This brief lists the different reports but does not substitute this figure for formal design documentation. Reservoir water level, flood-season limiting level, design flood level, spillway-crest elevation, and dam-crest elevation all have distinct engineering meanings.

Appendix 3. Downstream River-Recreation Safety Advisory

- Altered channel bed: High-velocity flooding may create deep holes, concealed channels, abrupt drops, and zones of strong current that cannot be judged visually.
- Submerged hazards: Embankment soil, trees, reinforcing steel, building debris, electrical wires, and household waste may create collision and entanglement hazards.
- Water-quality hazards: Floodwater may carry pathogens, decaying organic matter, pesticides, and other contaminants.
- Bank collapse: After floodwater scours the toe of a bank, the bank may still collapse suddenly during recession.
- Infrastructure hazards: Bridges, culverts, levees, and roadbeds may have structural damage that is not readily visible.

Swimming, wading, boating, fishing, and approaching damaged bridges, culverts, or steep banks should be avoided until safety assessment, debris removal, sediment cleanup, and water-quality testing have been completed.